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Enhanced Single-Trip Intelligent Completion Installation with ESP and a Downhole Wet-Mate

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Abstract

A cost-effective approach to improving reservoir management and optimizing production involves the use of electrical submersible pumps (ESP) with the incorporation of intelligent completion systems. By employing hydraulic downhole wet-mate connector technology, the operator is able to leverage the benefits of intelligent completions for managing multizone production. This approach significantly reduced intervention costs and mitigated the operational challenges associated with ESP replacements. Often, the operator is faced with significant challenges in managing multizone production in wells equipped with Electrical Submersible Pumps (ESPs) and the complexities of ESP replacements. To address these issues, Intelligent Completions were integrated with a hydraulic downhole wet-mate connector. This integration enabled the retrieval of the upper completion, including underperforming ESPs or safety valves, without the need to kill the well or remove Intelligent Completion system components such as zonal isolation packers and inflow control valves (ICVs).

This paper explores further optimizations that streamline the deployment process, allowing it to be completed in a single trip without disconnections. This improvement has substantial positive impacts on health, safety, and environment (HSE), reduces formation exposure, simplifies deployment complexity, and lowers capital expenditures (CAPEX).

The innovative single-trip intelligent completion system with ESP and downhole wet-mate connector streamlines the installation process by eliminating the need for ICVs and zonal feedthrough isolation packers to be run on drill pipe. All completion components are installed in a single trip without disconnecting the wet-mate during well completion. The lower completion zonal feed-through isolation packers are set using a surface hydraulic pump via the packer setting control line, followed by setting the ESP packer by applying tubing pressure against a slickline plug.

This system was successfully deployed in the Middle East, resulting in a significant time saving of 34 hours of rig time. The complexity and HSE risks were mitigated by removing the need to pull the drill pipe with control lines and large clamps, and by isolating the formation during tubing hanger landing. All well objectives were achieved, with all equipment deployed successfully in a single trip, confirming the integrity of the packers and the successful functioning of the ICVs.

This paper, thus, delves into the innovative implementation of an enhanced single-trip intelligent completion system featuring Electric Submersible Pumps (ESP) and downhole wet-mate technology. It also emphasizes the use of control-line set zonal feed-through isolation packers, showcasing the novel aspects and benefits of this approach.

Introduction

Mature oil and gas fields present complex reservoir challenges, necessitating sophisticated well drilling and completion techniques. These challenges include variations in the reservoir rock (heterogeneity), the potential for unwanted water or gas production (breakthrough), thin layers of oil and gas, and changes in rock types along the wellbore. (Arackakudiyil 2025).To overcome these issues and maintain good production rates, a combination of advanced technologies is typically employed. These include drilling multiple well branches from a single wellbore (multilaterals), using devices to control fluid flow into the well (inflow control devices or ICDs), employing electric submersible pumps (ESPs) for artificial lift, and utilizing intelligent completions (ICs) for remote monitoring and control. These technologies have a long track record of successful application, with ESPs being used for over 80 years and ICs for roughly 30 years(Mohanna 2011).

Combining these technologies allows for customized well completions that effectively address the reservoir complexities and ensure that ICs and ESPs can work together smoothly. Integrating intelligent completions with ESPs offers several advantages for boosting production and recovery while lowering lifting costs (Dossary 2016). By controlling water production zones, ESP energy can be focused on lifting oil, enabling greater pressure reduction in less productive areas. In long horizontal wells, flow is managed to delay early water breakthrough. Furthermore, the size of ESPs and gas handling equipment can be optimized to better match the oil lifting requirements. This integration also reduces wear and tear caused by erratic flow (slugging) and gas in the fluid, and it allows the well to be safely shut in at the reservoir level during ESP maintenance, maintaining well control and minimizing damage to the formation.

Intelligent Completions empower operators to monitor, manage, and optimize well and field performance(Arackakudiyil 2025). Initially adopted in subsea wells to produce from multiple zones without the need for costly interventions, the high initial investment in ICs was justified by the savings on these interventions. ICs provide remote control over downhole valves, allowing adjustments to production rates without physical intervention. This technology enables real-time reservoir monitoring and production adjustments. If unwanted fluids like water or gas enter one branch of a multilateral well, that branch can be shut off, allowing uninterrupted production from other branches. This leads to better production optimization, increased overall recovery, and fewer future interventions. Additionally, ICs improve the process of cleaning up the well after drilling by selectively opening different sections. Today, ICs are used in both onshore and offshore environments to manage complex multilateral wells, control water production, and monitor the integrity of the wellbore. They offer a cost-effective way to drain reservoirs, increase and sustain production, minimize the production of unwanted fluids, and improve overall reservoir recovery (Joubran 2018).

Globally, less than 10% of oil wells flow naturally; the majority rely on artificial lift methods, often employing ESPs (Mohanna 2011). An ESP consists of a multistage pump and an electric motor, connected to surface control equipment via a protected cable. The pump stages, each containing a rotating impeller and a stationary diffuser, are driven by a three-phase electric motor. Traditionally, ESPs are used to enhance production in wells with low reservoir pressure or high water content. They are also sometimes used in naturally flowing wells to manage reservoir energy and help maintain pressure.

The integration of ESPs with Intelligent Completions often involves a Hydraulic Wet-Mate Connection. Hydraulic control lines run alongside the ESP to operate the intelligent control valves (ICVs). This wet-mate system allows the ESP to be removed and reinstalled without having to remove the IC, resulting in significant time and cost savings. Furthermore, because the existing lower completion components remain in place and available, the already-installed interval control valves can provide complete isolation of the reservoir during workover operations. This helps to reduce health, safety, and environmental risks, save time, and optimize operational costs (Arackakudiyil 2025).

This paper discusses enhancements in well operations for the deployment of intelligent completions with ESPs, aimed at reducing risk and improving efficiency.

Completion Designs

For wells utilizing Intelligent Completions, a well-thought-out completion design is essential for achieving the best possible performance. This design includes both the upper and lower sections of the well's completion. The lower completion typically incorporates Interval Control Valves and packers that create seals between different production zones, allowing for the management of production from multiple zones. The upper completion mainly consists of the Electrical Submersible Pump, which provides artificial lift to bring fluids to the surface.

A vital element of this design is the Hydraulic Wet-Mate Connector, positioned between the upper and lower completions. This connector allows the ESP to be retrieved and reinstalled without affecting the components of the lower completion, such as the ICVs or zonal isolation packers. This capability means that if the ESP needs replacement, it can be done while keeping the Intelligent Completion system intact, thus minimizing intervention time and associated costs (Arackakudiyil 2025).

Early applications, such as those in Operators Smart Wells, typically involved a two-trip system for installing the completion. However, technological advancements, including the integration of control line set packers, have led to a more efficient single-trip approach. This single-trip method streamlines the installation process and reduces the risks associated with multiple trips into the well, resulting in reported rig time savings of up to 27 hours. Further innovations have produced an "Enhanced Single-Trip" solution, which eliminates the need for a separate space-out operation and can further reduce rig time by an additional 12 hours, bringing the total rig time savings to 39 hours. These improvements significantly boost operational efficiency, lower risks, and enhance overall cost-effectiveness.

Two trip systems

The conventional two-trip completion method involves two separate runs into the well. The initial trip focuses on installing the lower completion, which includes setting the packers and other components required to isolate different production zones. This deployment is achieved using drill pipe to reach the target depth (Almasri 2023). After the lower completion is in place, the drill pipe is removed from the well. A second trip is then made to install the upper completion, which comprises the ESP and its associated tubing hanger as shown in Fig 1.

Although this two-trip system is a functional approach, it generally demands more time and resources compared to more modern techniques. Typically, the first trip involves installing the lower intelligent completion, where packers and ICVs are set using drill pipe. The second trip then deploys the upper completion, including the ESP, which is connected to the lower completion before setting the tubing hanger. While still utilized in some cases, this method has largely been replaced by more efficient single-trip alternatives.

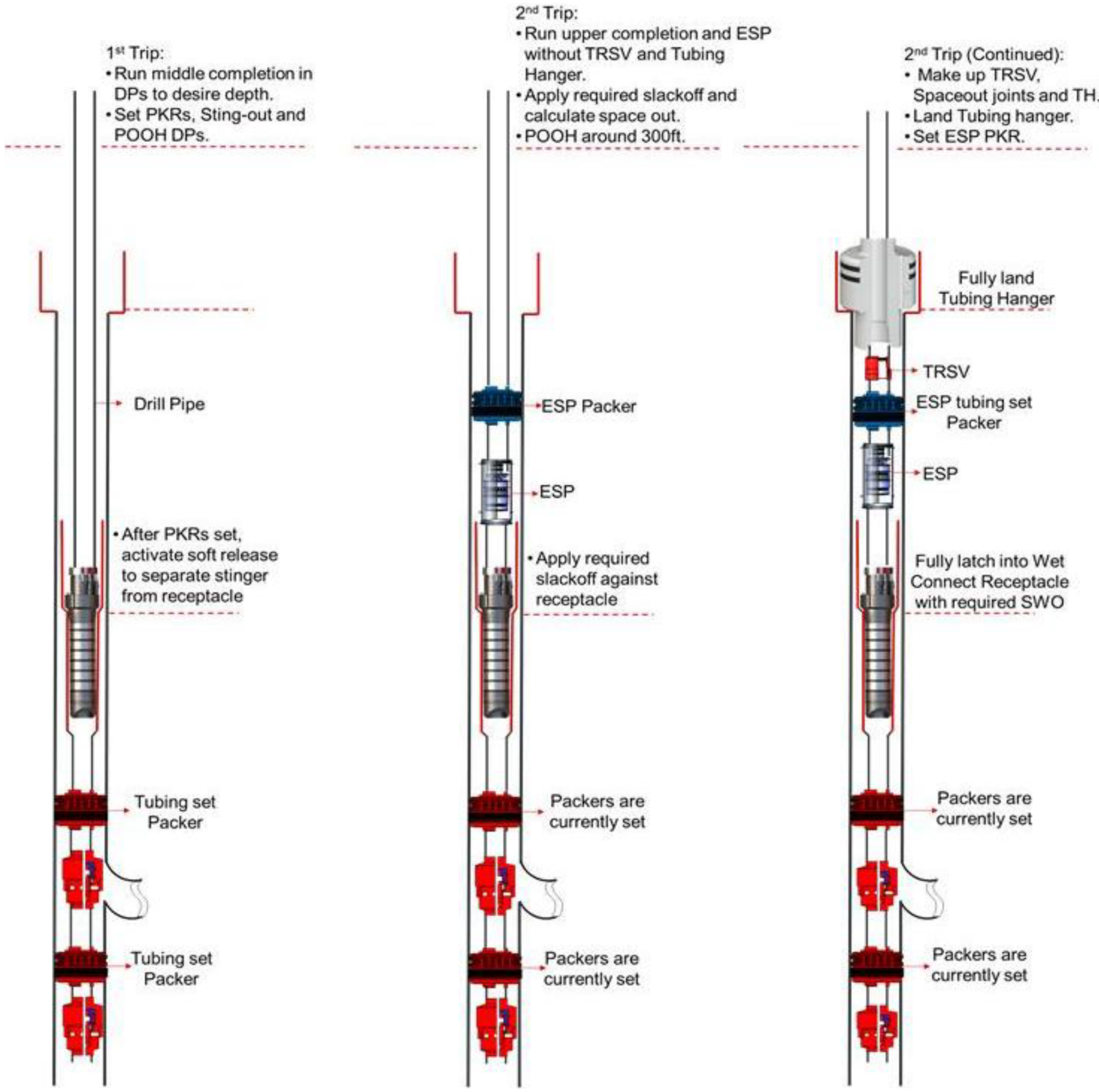

Figure 1—Summary of the Two-Trip System for Hydraulic Wet-Mate Connection Installation

Single trip systems

The development of control line set packer technology paved the way for the single-trip completion system, which enables the installation of both the upper and lower completions in a single run into the well. This approach offers substantial operational benefits, such as decreased rig time, reduced potential hazards, and simpler logistical requirements. By combining the installation of the intermediate and upper Smart/ ESP completions into one trip, the process is streamlined, and the risks associated with multiple trips are minimized (AlRaml 2024).

In a typical single-trip operation, the packers in the intermediate completion are set using specialized control line set packers. Following the setting of these packers, the upper completion, including the ESP, is integrated and secured without the need for additional tubing hangers or subsurface safety valve (TRSV) setups. This simplified procedure can save up to 27 hours of rig time compared to the conventional two-trip method, making it a highly efficient solution for completing wells (Arackakudiyil 2025).

Enhanced Single-Trip Operation:

The Enhanced Single-Trip system is a further refinement of the single-trip completion method. This advanced approach eliminates the need for a separate space-out operation, which is usually required to ensure proper alignment when installing the upper and lower completions. By calculating the required spacing for the entire completion string beforehand, this enhanced system not only cuts down on rig time but also reduces the risk of wellbore fluids and debris contaminating the hydraulic connections of the Wet-mate system.

In an Enhanced Single-Trip operation, the intermediate and upper IC/ESP completions are run into the well together with the tubing hanger that includes the subsurface safety valve (TRSV) as shown in Fig 2. Once the assembly reaches the desired depth, the tubing hanger is positioned just above its final landing point, allowing the packers in the intermediate completion to be set. After the packers are set, the tubing hanger is fully landed, and the ESP packer is then set. This improved process removes the need for manual space-out measurements and significantly lowers operational risk, providing a safer and more efficient installation.

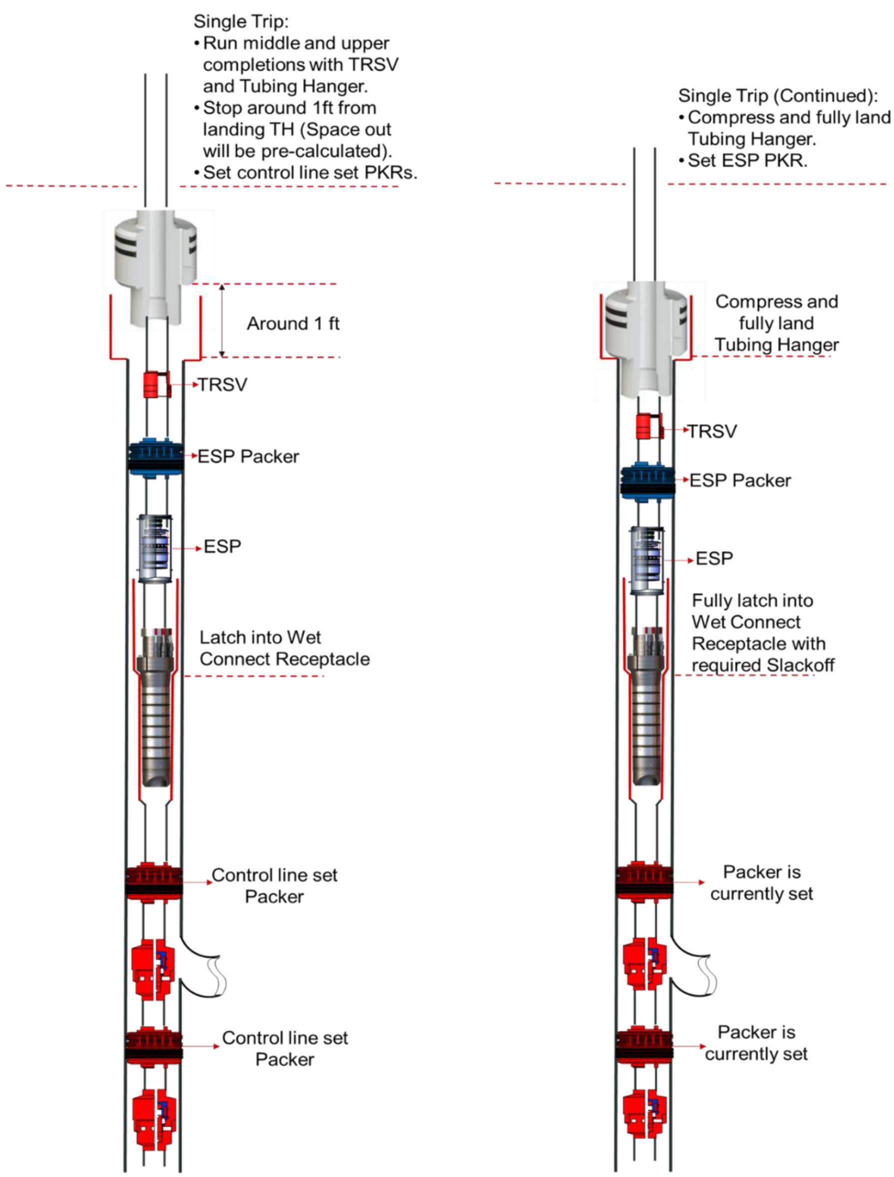

Figure 2—Summary of the Enhanced Single Trip System for Hydraulic Wet-Mate Connection Installation

Comprehensive modeling is considered vital to ensure the integrity of the completion in each Enhanced Single-Trip job. Thorough analysis including the space-out calculation is involved to guarantee that all components fit seamlessly throughout the well's lifespan. Each step is carefully planned to avoid potential problems and uphold the well's reliability. The meticulousness of this modeling is seen as key to the success of Enhanced Single-Trip operations.

The main advantages of the Enhanced Single-Trip system include an approximate 12-hour reduction in rig time, building upon the 27-hour savings of the standard single-trip method. Furthermore, this solution prevents wellbore debris from contaminating the hydraulic areas of the wet-mate connector, thus preserving the system's integrity. By optimizing both time efficiency and reliability, the Enhanced Single-Trip system offers a highly effective and cost-efficient solution for completing wells.

Workover and ESP replacement

Replacing an Electrical Submersible Pump in a well with Intelligent Completions can be a complicated task, especially with multiple packers and downhole components like hydraulically controlled interval control valves. Given that ESPs typically last around five years, regular replacement or maintenance is necessary. Traditionally, this would mean pulling the entire Intelligent Completion system out of the well, a time-consuming and expensive procedure.

However, incorporating a downhole Wet-Mate tool offers a significant advantage in this situation. This tool allows the ESP to be replaced or serviced without disturbing the intelligent completion system. This is particularly useful in wells producing from multiple zones with several ICVs, as it enables ESP replacement without the need to remove and reinstall the entire completion string.

The ability to replace the ESP without disrupting the Intelligent Completions System provides several key benefits:

Reduced Risk and Complexity

By eliminating the need to retrieve the entire intelligent completion, the process becomes less risky and more straightforward. Avoiding the removal and reinstallation of multiple packers, control lines, and related equipment reduces the chance of damage or operational failures.

Simplified Workover Process

The stab-in and stab-out feature of the Wet-Mate tool allows for ESP replacement without rotation and the need for excessive pulling force, simplifying the workover process. This reduces the time required for intervention and minimizes the complexity of ESP replacement, ultimately lowering operational costs.

Minimized Equipment Handling

Since the Intelligent Completion remains in place, there is no need to handle or reinstall critical downhole components such as ICVs and packers. This reduction in handling steps lowers the potential for errors and minimizes non-productive time.

Cost savings through Integration of Intelligent Completions and ESP System

Combining intelligent completion technology, such as the Hydraulic Wet-Mate system, with Electrical Submersible Pump systems yields significant benefits, leading to substantial cost savings, increased oil recovery, and more effective reservoir management. This integration optimizes system performance, simplifies operations, and allows for the use of smaller equipment, thereby lowering the costs associated with lifting fluids.

Here are the main ways cost savings are achieved:

- **Optimized Energy Use:** By selectively restricting or shutting off zones producing water, the ESP system focuses its energy on lifting oil, improving efficiency and reducing both energy consumption and the expenses related to handling water.

- **Greater Flexibility in Managing Pressure Reduction:** Intelligent completions with Inflow Control Valves offer more flexibility in managing pressure differences across different reservoir zones. This allows operators to maximize production from less productive or lower-pressure areas, enhancing oil recovery, reducing the need for expensive interventions, and improving the long-term economic viability of the well.
- **Improved Well Control and Protection:** Intelligent completions enhance well control by enabling operators to shut in the well at the reservoir level when retrieving the ESP. This reduces the risk of damage to the formation and improves the well's integrity, especially during interventions. Better well control minimizes the cost and complexity of interventions, reduces non-productive time, and lowers the frequency of workovers, all of which lead to significant cost reductions.
- **Reduced Operational Wear and Tear:** Intelligent completions lower the risk of erratic flow (slugging), gas entering the pump, and other production issues that can accelerate wear on the ESP system. By optimizing flow control, these systems extend the operational life of the ESP, minimize the need for maintenance, lower repair costs, and contribute to overall cost savings throughout the well's lifespan.

HSE benefits

Combining the Hydraulic Wet-Mate system with intelligent completions and Electrical Submersible Pump systems offers substantial health, safety, and environmental (HSE) advantages. This integration enhances operational safety, minimizes environmental risks, and preserves the integrity of the well. By decreasing non-productive time, simplifying workover processes, and reducing personnel exposure to high-risk activities, this technology provides a comprehensive solution for more efficient and safer reservoir management. Furthermore, the ability to replace or service ESPs without disturbing critical downhole equipment, such as interval control valves or zonal isolation packers, ensures both safety and operational efficiency. This approach not only delivers significant cost savings but also upholds the highest HSE standards throughout the well's operational life.

Conclusion

To lower upfront costs (CAPEX) and simplify the complexities of workovers needed for ESP replacements, a downhole wet mate system has been integrated into intelligent completions. This innovative technique, utilizing the Hydraulic Wet-Mate Connection, has been successfully implemented in over 60 wells. Further advancements have streamlined the deployment process, enabling single-trip installations. This improvement has significantly boosted health, safety, and environmental standards, reduced the risk of formation damage, simplified deployment procedures, and lowered CAPEX.

These implementations have showcased the effective combination of a reliable completion system that integrates intelligent completions with ESPs for artificial lift. When producing from multiple reservoirs at the same time, this integrated system minimizes the risk of fluid movement between different reservoirs and maximizes oil production rates from all well branches and reservoirs. The success of these implementations is a result of careful planning and execution at every stage. The new single-trip system design has not only minimized potential hazards but also improved performance through seamless operation and the inclusion of robust contingency plans. The industry continues to advance in reducing operational durations and mitigating associated risks.

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Nomenclature

ESP	Electrical Submersible Pump
ICV	Interval Control Valves
ICD	Inflow Control Device
IC	Intelligent Completion
MRC	Maximum Reservoir Contact
RIH	Run in Hole
NPT	Non-Productive Time

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